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(i) Logical bitwise operators:

&	Bitwise AND	(Binary AND)
|	Bitwise OR	(Binary OR)
^	Bitwise Exclusive OR	(Binary XOR)

- Bitwise AND (&) will return 1 (True)if both the operands are having the value 1 (True); Otherwise, it will return 0 (False)
- Bitwise OR (|) will return 1 (True) if any one of the operands is having a value 1 (True); It returns 0 (False) if both the operands are having the value 0 (False)
- Bitwise XOR (^) will return 1 (True) if only one of the operand is having a value 1 (True). If both are True or both are False, it will return 0 (False).

Truth table for bitwise operators (AND, OR, XOR)

A	B	A & B	A | B	A ^ B
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

Example:

If a = 65, b=15

Equivalent binary values of 65 = 0100 0001; 15 = 0000 1111

<table>
<tr><th>Operator</th><th>Operation</th><th>Result</th></tr>
<tr><td>&</td><td>a & b</td><td>
<table>
<tr><td>a</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>b</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>a & b</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
</table>
(a&b) = 0000 0001$_2$= 1$_{10}$
</td></tr>
<tr><td>|</td><td>a | b</td><td>
<table>
<tr><td>a</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>b</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>a | b</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
(a|b) = 01001111$_2$= 79$_{10}$
</td></tr>
<tr><td>^</td><td>a ^ b</td><td>
<table>
<tr><td>a</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>b</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>a ^ b</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr>
</table>
(a^b) = 0100 1110$_2$= 78$_{10}$
</td></tr>
</table>